$$Hd(e^{j\Omega}) = \begin{cases} A; & |\Omega| < \Omega_0 \\ 0; & \text{otro caso} \end{cases}$$

$$Hc(j\omega) = \begin{cases} 1; & |\omega| \geq 2\pi 2000 \\ 0; & \text{otro} \end{cases}$$

$W = 2\pi 4000$; $f_s = 10.000\,Hz$

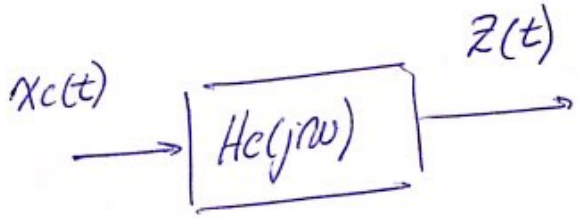
$\hookrightarrow \omega_s = 2\pi 10.000$

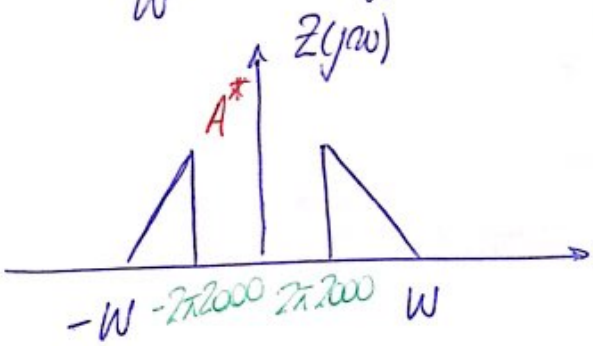
$\omega_s \geq 2W$

$2\pi 10.000 \geq 2\pi 8000$ ✓ → NO HAY ALIASING!

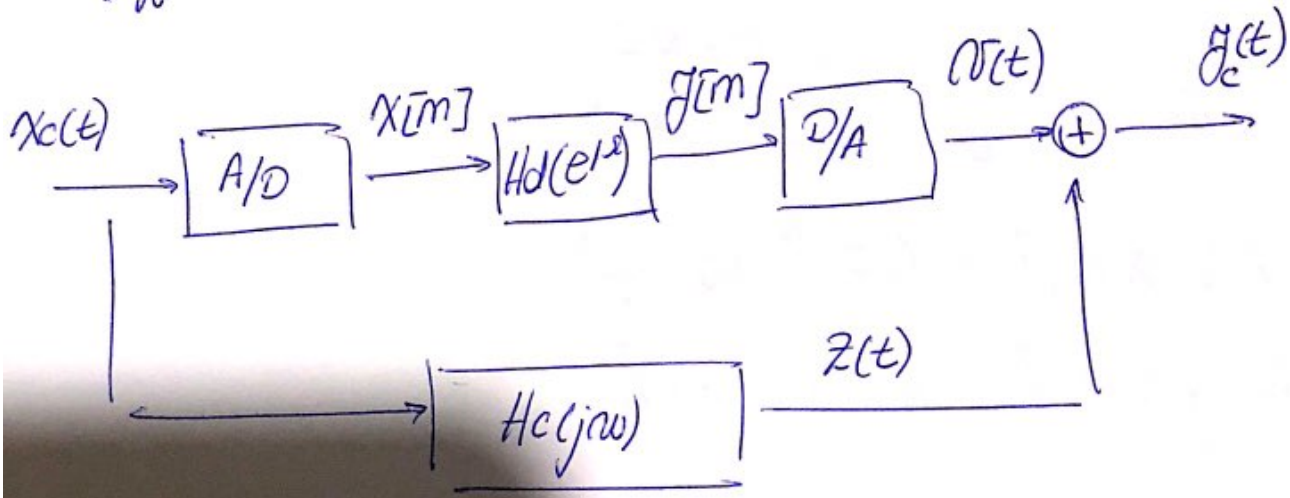
PROPONGO:



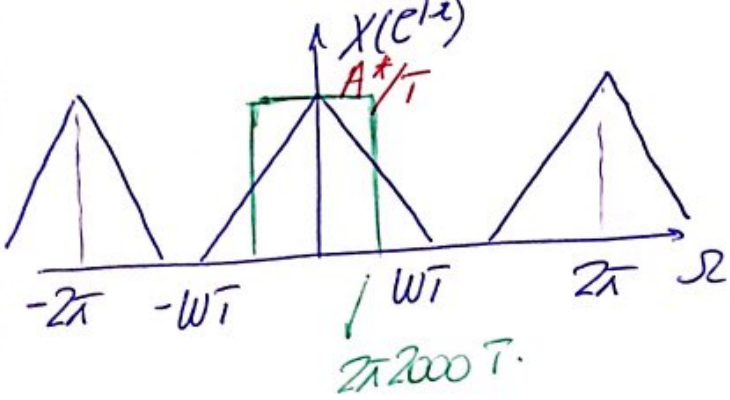




VIENDO Z(jw) PODEMOS DECIR QUE U(jw) TIENE QUE SER LO QUE LE FALTA A Z(jw) PARA SER IGUAL A Xc(jw).



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$$\Rightarrow \omega_0 = 2\pi 2000 \cdot T \; ; \; T = \frac{1}{10.000} = 0{,}1\text{ms}$$

$$\boxed{\omega_0 = \frac{2}{5}\pi}$$

SI CONSIDERO QUE

$$H_r(j\omega) = \begin{cases} T & ; |\omega| \leq 2\pi 2000 \cdot \\ 0 & ; \text{otro caso} \cdot \end{cases}$$

$\Rightarrow$ A TIENE QUE SER / $AA^* = A^*$

$$\Rightarrow \boxed{A = 1 \cdot}$$

3) $x[n]$ TIENE LONGITUD N CON TRANSFORMADA $X(z)$

a) $$x_1[n] = \begin{cases} x[n] & 0 \leq n \leq N-1 \\ 0 & N \leq n \leq 2N-1 \end{cases}$$

$$\text{DFT}\{x[n]\} = X(e^{j\omega})\Big|_{\omega = \frac{2\pi}{N}K}$$

$$\text{DFT}\{x[n]\} = X(z)\Big|_{z = e^{j\frac{2\pi}{N}K}}$$

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RAZIEN $\text{LONG}\{x_1[m]\} = 2N$

$$\Rightarrow X_1[K] = X(z)\Big|_{z=e^{j\frac{\pi}{N}K}}$$

b) $x_2[m] = x[m] + x[m-N]$; $\text{LONG}\{x_2[m]\} = 2N$

$$X_2(z) = X(z) + z^{-N}X(z)$$

$$\Rightarrow X_2[K] = X(z) + z^{-N}X(z)\Big|_{z=e^{j\frac{\pi}{N}K}}$$

$$X_2[K] = X(z) + e^{j\frac{\pi}{N}K(-N)}X(z)\Big|_{z=e^{j\frac{\pi}{N}K}}$$

$$X_2[K] = X(z) + e^{-j\pi K}X(z)\Big|_{z=e^{j\frac{\pi}{N}K}}$$

$$X_2[K] = X(z) + (-1)^n X(z)\Big|_{z=e^{j\frac{\pi}{N}K}}$$

$$\Rightarrow X_2[K] = \begin{cases} 2X(z); & z=e^{j\frac{\pi}{N}K} \cdot K = \text{PAR} \\ 0 ; & z=e^{j\frac{\pi}{N}K} \cdot K = \text{IMPAR} \end{cases}$$

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